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Data Hiding Method for English Scripts using Ligature Characters Unicode

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Abstract

Data hiding has appeared as an important research field to resolve the problems in network security, and secure communications through public and private channels. Text steganography is most challenging because of the presence of very less redundant information in text documents as compared to the images and audio. In this paper a novel method is proposed for data hiding in English Scripts using ligature characters Unicode (where characters are generally joined together). In this method, replace two ASCII codes of ligature characters by one Unicode of same ligature characters based on data which is hiding. This method has a hiding capacity based on specific characters in each document. As well as have good perceptual transparency and no changes in original text.

Keywords: Text Hiding, Unicode Standard, Ligature Characters.

1. Introduction

Steganography is a process that involves hiding a message in an proper carrier such as image, text or voice. It is means "covered or hidden writing"in Greek. For secret communication purpose, steganography protect both messages and its destination parties such that the authenticated receiver only knows existence of the messages that carrier sent. A large amount of the works being done by a variety of research in steganography [1]. Steganography refers to the art and science of hiding secret information in some other media. Secret message is the information to be hided and the cover document is the medium in which the information is hided. The cover document containing hidden message is called stego-document. The algorithms employed for hiding the message in the cover medium at the sender end and extracting the hidden message from the stego-document at the receiver end is called stego system. [2].

In this paper, some ligatures characters have high frequency in English scripts. Thus, high advantages was obtained for using Unicode of these ligature characters in the process of concealment.

The rest of the paper is organized as follows. Section 2 presents the previous works that have been done in the field of text steganography. Section 3 presents the proposed method in detail. Section 4 and 5 for results and concludes the paper.

2. Previous Works on Text Steganography

There are a number of researches had already explored in new steganographic techniques in texts, such as white spaces[3], Synonyms[4], Word Shifting[5], and Line shifting [6]. This paper focused on researches which used Unicode in data hiding, M. H. Shirali-Shahreza, and Mohammad Shirali-Shahreza proposed new method for hiding information in Persian and Arabic Unicode texts[7]. Also, they proposed another method for hiding data in Persian (Farsi) and Arabic texts. Based on characters of « ی"and « ک"have the same shape but different codes[8]. Lip Yee Por and *et al*. proposed method based on Unicode space characters with respect to embedding efficiency[9].

3. Proposed Text Steganography Method

In this paper, a new method was presented for text Steganography in English Scripts using Unicode of ligatures characters. In the following section, Unicode Standard was declared, then explain the Ligatures Characters briefly, and lastly, the suggested method explained in details.

3.1. Unicode Standard

The problems of growing information exchange, is the introduction of a character code which gives every character of every language a unique number. The range of possible numbers is set adequately high to cover all the current and future needs of all languages. This number does not depend on the language used in the text, the font used to display the character, the software, the operating system, or the device[10]. The Unicode consortium enables people around the world to use computers in any language[8].

Unicode Standard is the international character-encoding standard used for presenting the texts to process by computers. All the characters used in writing of the world languages can be encoded by the Unicode standard uses the 16-bit encoding which provides space for 65000 characters in different moulds such as symbols, numbers, letters, and a great number of current characters in different languages of the world[10].

There were different encoding systems which have conflict with one another. That is, two encodings can use different numbers for the same character, or use the same number for two different characters, or When data is passed between different encodings or platforms, that data always runs the risk of corruption; therefore, many different encodings for any given computer (especially servers) needed to provide. Unicode provides a unique number for every character, which is independent from the platform, the program, and the language[10].

3.2. Ligature Characters

A ligature is a visible presentation of two or more characters as a unit. The origin of ligatures is in cursive handwriting, where characters are generally joined together. For printing, ligature types were produced to solve problems in both the visual appearance and in the mechanics of printing. Text does not look good if, for example, the upper part of the "f"is close to dot of the "i"as in "fi,"so it can be better to create a type that fuses the two characters together[11]. There are some constructs specifically related to ligatures in Unicode, For example, there is Latin small ligature "fi"(U+FB00), which is defined as equivalent compatibility-wise to "f"followed by "i"[11].

In general, Ligatures characters (109 characters) in Unicode Standard Version 6.2 in English scripts and other scripts are tabulated in Table (1).

Table 1: Ligatures Characters in Unicode Standard Version 6.2

	Ligatures Characters									
	Char.	**Code**	**Char.**	**Code**	**Char.**	**Code**	**Char.**	**Code**	**Char.**	**Code**
Latin Extended-A (0100-017F)	IJ	0132	ij	0133						
Latin Extended-B (0180-024F)	DZ lj	01F1 01C9	Dz NJ	01F2 01CA	dz Nj	01F3 01CB	LJ nj	01C7 01CC	Lj	01C8
Latin Extended Additional (1E00-1E31)	IL	1EFA								
Currency Symbols (20A0-20CF)	Pts	20A7	Rs	20A8						
Letterlike Symbols (2100-214F)	SM	2120	TM	2122						
Number Form (2150-218F)	ii VI	2171 2165	II XI	2161 216A	iv xi	2173 217A	vi IX	2175 2168	IV ix	2163 2178
Enclosed CJK Letters and Month (3200-32FF)	PTE	3250	Hg	32CC	erg	32CD	eV	32CE	LTD	32CF
Alphabetic Presentation Form (FB00-FB4F)	ff ft	FB00 FB05	fi	FB01	fl	FB02	ffi	FB03	ffl	FB04
CJK Compatibility (3300-33FF)	hPa pc mA cal kg THz km rad nV nW cc in log pH Wb	3371 3376 3383 3388 338F 3394 339E 33AD 33B5 33BB 33C4 33CC 33D2 33D7 33DD	da dm kA kcal Hz fm Pa ps mV mW cd KM lx PPM gal	3372 3377 3384 3389 3390 3399 33A9 33B0 33B7 33BD 33C5 33CE 33D3 33D9 33FF	AU IU KB pF kHz nm kPa ns kV kW dB kt mb PR	3373 337A 3385 338A 3391 339A 33AA 33B1 33B8 33BE 33C8 33CF 33D4 33DA	bar pA MB nF MHz mm MPa ms MV MW Gy lm mil sr	3374 3380 3386 338B 3392 339C 33AB 33B3 33B9 33BF 33C9 33D0 33D5 33DB	oV nA GB mg GHz cm GPa pV pW Bq ha ln mol Sv	3375 3381 3387 338E 3393 339D 33AC 33B4 33BA 33C3 33CA 33D1 33D6 33DC

3.3. Proposed Method Explanation

In this paper, new method for text steganography based on Unicode of ligatures characters. Some important parameters must be regards to fulfill main goal of steganography design:

1. Glyphs of ligature characters in English scripts.
2. Appearance or not in English scripts.
3. Possibility changes to another glyphs.
4. Frequencies of ligature characters in English scripts.

Most ligature characters have glyphs different from stego-doc. glyph, such as "da",nm", "iv", … etc. which have "MS Mincho"font type; others have no glyphs(no appearance) in English scripts, or their appearance as (disorganized), such as "LJ"no appearance; and others have font as same as document font type such as "fi", as well as can be changed it.

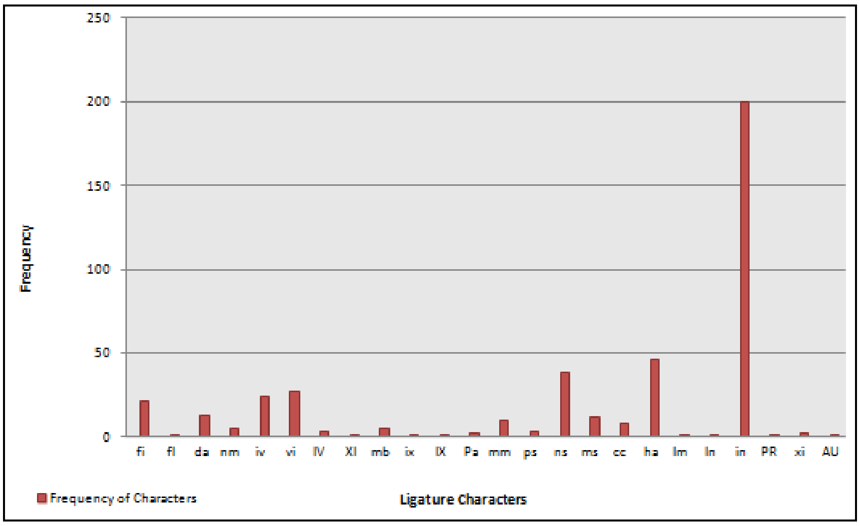
Fortunately, most ligature characters glyphs can be changed to specific glyph, and the frequencies of them very high. Figure(1) explains frequency of ligature characters in many English scripts files in different sizes. To overcome parameters above, and obtain perceptual transparency, the types of fonts can be used {"MS Mincho", "MS Gothic", "Batang","BatangChe"} in data hiding of proposed method (which is most similar to ligature characters glyphs), and some ligature characters was ignored, see table(2).

Table 2: Filtered Ligatures Characters

Filtered Ligature Characters									
Char.	**Freq.**	**Glyph**	**Font**	**Changed**	**Char.**	**Freq.**	**Glyph**	**Font**	**Changed**
fi	H	fi	Doc. font	yes	mm	M	mm	MS Mincho	yes
fl	M	fl	Doc. font	yes	ps	M	Ps	MS Mincho	yes
da	H	da	MS Mincho	yes	ns	H	ns	MS Mincho	yes
nm	M	nm	MS Mincho	yes	ms	M	ms	MS Mincho	yes
iv	H	iv	MS Mincho	yes	cc	M	cc	MS Mincho	yes
vi	H	vi	MS Mincho	yes	ha	VH	ha	MS Mincho	yes
IV	M	IV	MS Mincho	yes	lm	L	lm	MS Mincho	yes
XI	L	XI	MS Mincho	no	ln	L	ln	MS Mincho	no
mb	M	mb	MS Mincho	no	in	VH	in	MS Mincho	no
ix	M	ix	MS Mincho	yes	PR	L	PR	MS Mincho	no
IX	L	IX	MS Mincho	yes	xi	M	xi	MS Mincho	no
Pa	L	Pa	MS Mincho	yes	AU	L	AIJ	MS Mincho	no

H: High, M:media, L:Low, VH: Very High

Figure 1: Frequencies of Ligatures Characters

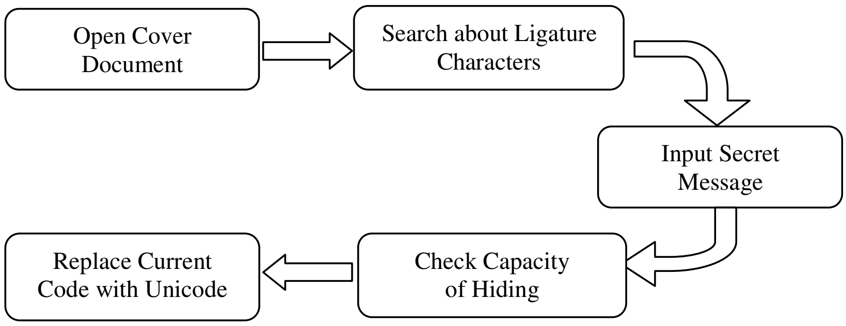


Now, in proposed method, two processes was implemented, hiding process, and extracting process. Hiding process explained in figure(2). Ligature characters must be located in document; Programmatically, these characters stored in one dimension array based on table (2), then embedding process implemented depending on secret message which can be hidden.

Checking capacity of hiding is a task to check if the document file have enough area to hide data (which must be adequate to hide secret message with two bytes represent message length). The message length embedded in beginning, and replacement can be done based on secret message as follows:

$$\text{If Secret Message} = \begin{cases} 0 & \text{No change} \\ 1 & \text{Replace with Unicode of Ligature Characters} \end{cases} \qquad (1)$$

Figure 2: Hiding Process



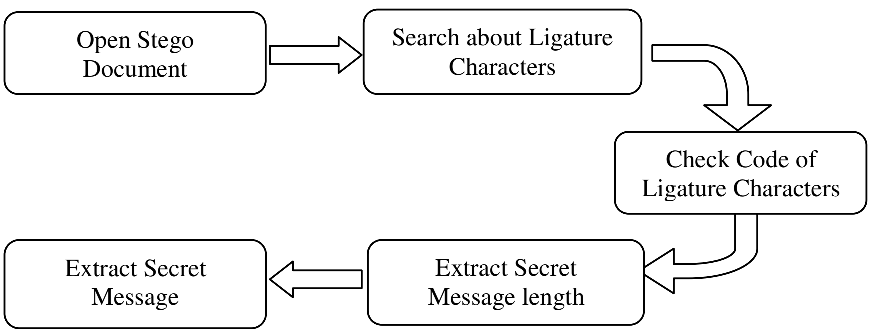
Hiding process (Embedding process) summarized in the following algorithm :

Embedding Algorithm
Input: Document file, and Secret Message.
Output: Stego file.
1. Open cover document. 2. Scan cover document to find Ligatures Characters, 3. Compute number of Ligatures Characters to check the capacity of hiding. 4. Get binary form of secret message. 5. For each symbol in secret message - if symbol = 0, then no change (ASCII code) , Else Replace by Unicode. 6. Return Stego document.

English Scripts written in (Latin Letters), take the range (U+0041-U+005A) for Uppercase Latin alphabet, and (U+0061-U+007A) for Lowercase Latin alphabet in hexadecimal. For example, there is Latin small ligature "in"which is defined as "i"(U+0069) followed by "n"(U+006E) can be no change if secret message is 0, but can be replaced by Unicode of ligature character "in"(U+33CC) when secret message is 1.

In another side, the extracting process is the opposite operation of the hiding process, see figure (3) which contain the steps of extracting process.

Figure 3: Extracting Process



Same procedure can be used to locate ligatures characters, then checking code of their by the following statement :

$$\text{If Code of Ligatures Characters} = \begin{cases} \text{In range}(U+0041\text{-}U+005A) \text{ or } (U+0061\text{-}U+007A) \\ \quad \text{or some special symbols then} \\ \quad \text{Secret message} = 0 \\ \text{Else} \quad \text{Secret message} = 1 \end{cases} \tag{2}$$

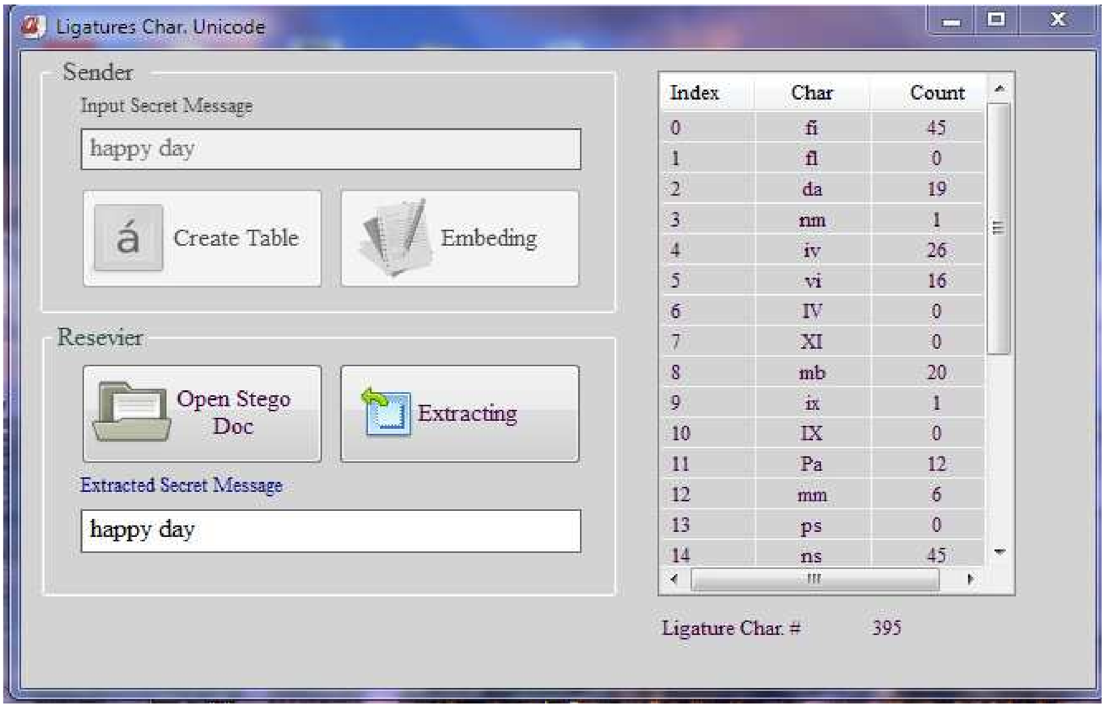
The first 16-bits represent the secret message length. Extracting process summarized in the following algorithm:

Extracting Algorithm
Input: Stego file.
Output: Secret Message.
1. Open Stego document. 2. Scan stego document to find Ligatures Characters, 3. Check code of ligatures characters based on equation (2) • if Code in Uppercase or Lowercase English Latin letters, Or special characters, then secret message=0 • Else secret message=1 4. Extract secret message length 5. Extract secret message(begin at 3rd byte).

4. Results

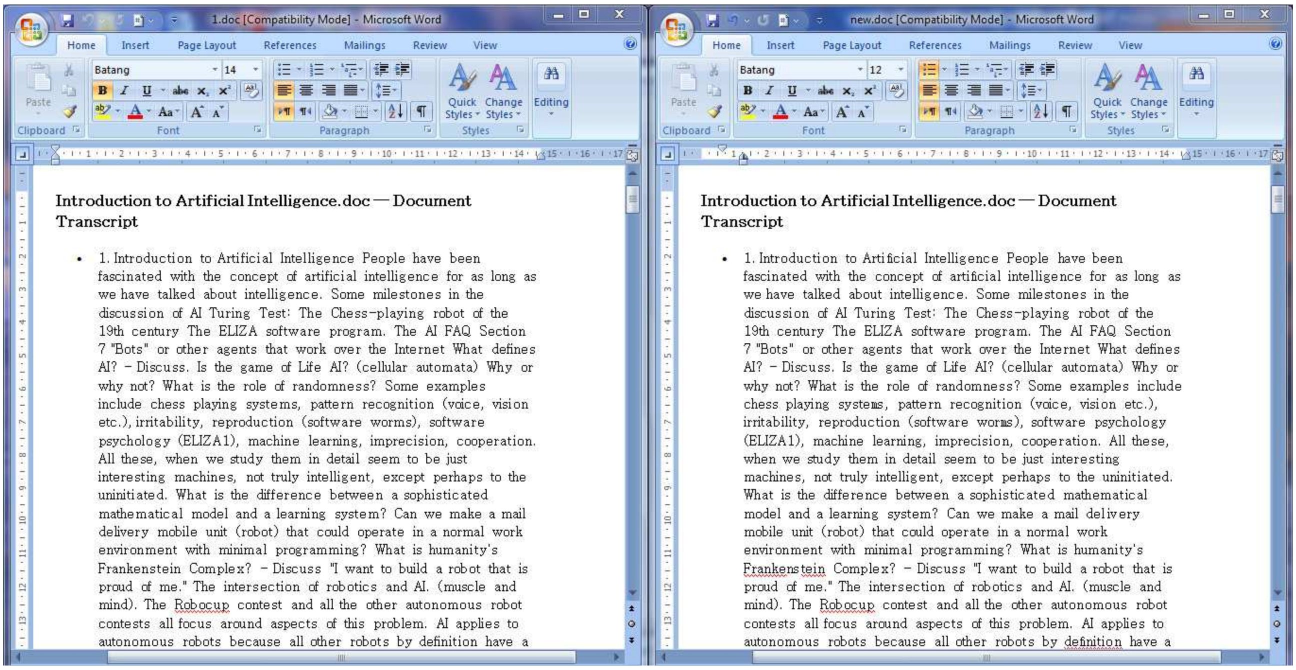
In this paper, the proposed method of data hiding is tested by taking different cover documents with different sizes in {"MS Mincho", "MS Gothic", "Batang","BatangChe"}font types and hiding the same secret message in some of them, see the corresponding GUI for the proposed method in figure (4).

Figure 4: GUI of Proposed Method



The payload of bits can be hidden based on ligature characters number. If the cover file contains 200 ligature characters, we can hiding 200 bits in it. Figure (5) represent cover document and stego document respectively.

Figure 5: Result of Proposed Method



a: Cover Document b: Stego Document

The results that are got from these experiments can be summarized in the Table.3. Note the average of size for Stego document (after hiding) is increased about (1.76%) from original size.

Table 3: Experimental Results of Proposed Method

Experiment #.	Ligature Char.# in Cover	Total Char.# in Cover	Ligature Char. Ratio in Cover (%)	Max.# of bits Can be Embedded in Cover	Ratio of Stego-Doc. Size increasing (%)
1	566	17043	4.06	566	8.75
2	395	13932	5.8	395	0.86
3	448	12115	3.6	448	0.28
4	210	5461	3.8	210	0.0
5	351	7118	4.9	351	---
6	1772	50807	3.4	1772	0.46
7	799	24880	3.2	799	0.7
8	977	27943	3.4	977	1.5
9	828	24964	3.3	828	3.38
10	1837	51486	3.5	1837	0.06
11	1326	44764	2.9	1326	0.4
12	111	3148	3.5	111	4.8
			Avrg. 3.78		**Avrg. 1.76**

5. Conclusion

In data hiding method, the main goals of steganography are(perceptual transparency, capacity, and robustness). Proposed method has a good perceptual transparency because the stego text is similar to the original text by use font types of cover document to be more similar and suitable to glyphs of ligature characters. The hiding capacity of proposed method is very high, depending on high frequency of ligature characters. In addition, this method is robust to digital copy-past operation, which means that copying and pasting the text between computer programs preserve hidden information, But it is vulnerable to retyping. However, it is efficient text steganography method by Unicode.

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